

Centralized Data Platform

AlloyTower Inc. | April 2026

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Project	Centralized Data Platform
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1. Project Objective

The objective of this project was to support the development of a centralized data platform for AlloyTower by preparing, organizing, and analysing the property dataset provided. Our team aimed to create a reliable foundation that would help answer the company's business questions, highlight market activity, and support future predictive modelling. To achieve this, we focused on understanding the structure of the data, improving its quality, and presenting insights that reflect the true capabilities of the dataset.

2. Data Cleaning Process

We began by examining the raw dataset and identifying issues that would affect accuracy and usability. These included blank rows, incorrect data types, inconsistent formatting, missing values, and duplicate property records. To address these problems, we:

- Removed unnecessary blank rows
- Corrected data types for IDs, dates, and numeric fields
- Standardized text fields for consistency
- Replaced missing unit values with "N/A"
- Eliminated duplicate property records

After cleaning, the dataset became more structured, consistent, and suitable for analysis, as shown in the before-and-after visuals below.

Before Cleaning

	A ^B _C property_id	1 ² ₃ id	A ^B _C full_address	A ^B _C street_address	A ^B _C unit
1			null		
2	P-100001		10001 1924 Oak Ave, Honolulu, HI 46048	1924 Oak Ave	
3			null		
4	P-100002		10002 3711 Harbor View Blvd, Pittsburgh, PA 87236	3711 Harbor View Blvd	
5			null		
6	P-100003		10003 8885 Park Way, Apt 21C, Kansas City, MO 59615	8885 Park Way	Apt 21C
7			null		
8	P-100004		10004 6165 Hillcrest Ave, Bronx, NY 37460	6165 Hillcrest Ave	
9			null		
10	P-100005		10005 625 Ridgewood Rd, Apt 14E, Tucson, AZ 62581	625 Ridgewood Rd	Apt 14E
11			null		
12	P-100006		10006 6030 Madison Ave, Grand Rapids, MI 28131	6030 Madison Ave	
13			null		
14	P-100007		10007 1976 Jackson Dr, Houston, TX 94012	1976 Jackson Dr	
15			null		
16	P-100008		10008 2746 Main St, Apt 22D, Cleveland, OH 88504	2746 Main St	Apt 22D
17			null		
18	P-100009		10009 2805 Adams St, Apt 19E, Indianapolis, IN 79163	2805 Adams St	Apt 19E
19			null		
20	P-100010		10010 9741 Madison Ave, Apt 26D, Tucson, AZ 10942	9741 Madison Ave	Apt 26D
21			null		
22	P-100011		10011 1645 River Blvd, Lawrenceville, GA 96374	1645 River Blvd	
23			null		

Figure 1: Raw dataset in Power Query — blank rows, null values, and type issues visible

After Cleaning

Queries [1] alloy_data						= Table.Distinct(#"Trimmed Text", {"property_id"})						Query Settings	
	A ^B _C property_id	A ^B _C id	A ^B _C full_address	A ^B _C street_address	A ^B _C unit	PROPERTIES						APPLIED STEPS	
1	P-100001	10001	1924 Oak Ave, Honolulu, HI 46048	1924 Oak Ave	N/A	Name	alloy_data					Source	
2	P-100002	10002	3711 Harbor View Blvd, Pittsburgh, PA 87236	3711 Harbor View Blvd	N/A	All Properties						Promoted Headers	
3	P-100003	10003	8885 Park Way, Apt 21C, Kansas City, MO 59615	8885 Park Way	Apt 21C							Changed Type	
4	P-100004	10004	6165 Hillcrest Ave, Bronx, NY 37460	6165 Hillcrest Ave	N/A							Removed Blank Ro...	
5	P-100005	10005	625 Ridgewood Rd, Apt 14E, Tucson, AZ 62581	625 Ridgewood Rd	Apt 14E							Changed Type1	
6	P-100006	10006	6030 Madison Ave, Grand Rapids, MI 28131	6030 Madison Ave	N/A							Replaced Value	
7	P-100007	10007	1976 Jackson Dr, Houston, TX 94012	1976 Jackson Dr	N/A							Changed Type2	
8	P-100008	10008	2746 Main St, Apt 22D, Cleveland, OH 88504	2746 Main St	Apt 22D							Trimmed Text	
9	P-100009	10009	2805 Adams St, Apt 19E, Indianapolis, IN 79163	2805 Adams St	Apt 19E							Removed Duplicates	
10	P-100010	10010	9741 Madison Ave, Apt 26D, Tucson, AZ 10942	9741 Madison Ave	Apt 26D								
11	P-100011	10011	1645 River Blvd, Lawrenceville, GA 96374	1645 River Blvd	N/A								
12	P-100012	10012	4663 Lakeshore Dr, Apt 6D, Decatur, GA 42742	4663 Lakeshore Dr	Apt 6D								
13	P-100013	10013	6496 Adams St, Apt 17E, Austin, TX 69638	6496 Adams St	Apt 17E								
14	P-100014	10014	1031 Birch St, Bronx, NY 20500	1031 Birch St	N/A								
15	P-100015	10015	5247 Monroe Blvd, Apt 29A, Naperville, IL 44814	5247 Monroe Blvd	Apt 29A								
16	P-100016	10016	6154 Heritage Ln, Apt 10E, Charlotte, NC 30676	6154 Heritage Ln	Apt 10E								
17	P-100017	10017	5717 Jefferson St, Apt 23D, Naperville, IL 93130	5717 Jefferson St	Apt 23D								
18	P-100018	10018	258 Park Way, Fort Wayne, IN 19862	258 Park Way	N/A								
19	P-100019	10019	5894 Willow Way, Apt 13A, Fort Lauderdale, FL 91351	5894 Willow Way	Apt 13A								
20	P-100020	10020	3744 Lincoln Rd, Apt 11A, Rockville, MD 70332	3744 Lincoln Rd	Apt 11A								
21	P-100021	10021	4379 Washington Ave, Orlando, FL 86099	4379 Washington Ave	N/A								

28 COLUMNS, 199+ ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 22:02

Figure 2: Cleaned dataset — duplicates removed, types corrected, blanks resolved (28 columns, 199+ rows)

3. Collaboration With the Data Scientist

After preparing the dataset, we reviewed it with the data scientist working with our team. During this discussion, it became clear that the dataset did not fully support all the original business questions. Several fields lacked clear definitions — particularly `assessor_id` and the additional `id` column — making it difficult to determine their purpose or how they should be used in modelling. The absence of a data dictionary also limited our ability to interpret certain fields confidently.

Because of these limitations, the data scientist recommended focusing on KPIs and visuals that describe the property itself, rather than financial or predictive metrics that the dataset could not reliably support. This allowed us to build a dashboard that reflects what the data can realistically show, while still supporting the data scientist's need for structured information to guide presumed predictive outcomes.

4. Team Alignment

These findings and limitations were shared with the Business Analyst and Project Managers to ensure alignment across the team. Everyone agreed that the analysis should focus on reliable, property-based insights that could still contribute to the broader goal of building a centralized data platform. This alignment ensured that expectations were realistic and that the final dashboard accurately represented the strengths and gaps within the dataset.

5. KPI Development

Based on the dataset's structure and the data scientist's guidance, the dashboard includes three KPIs that provide a clear overview of the dataset's scope and value:

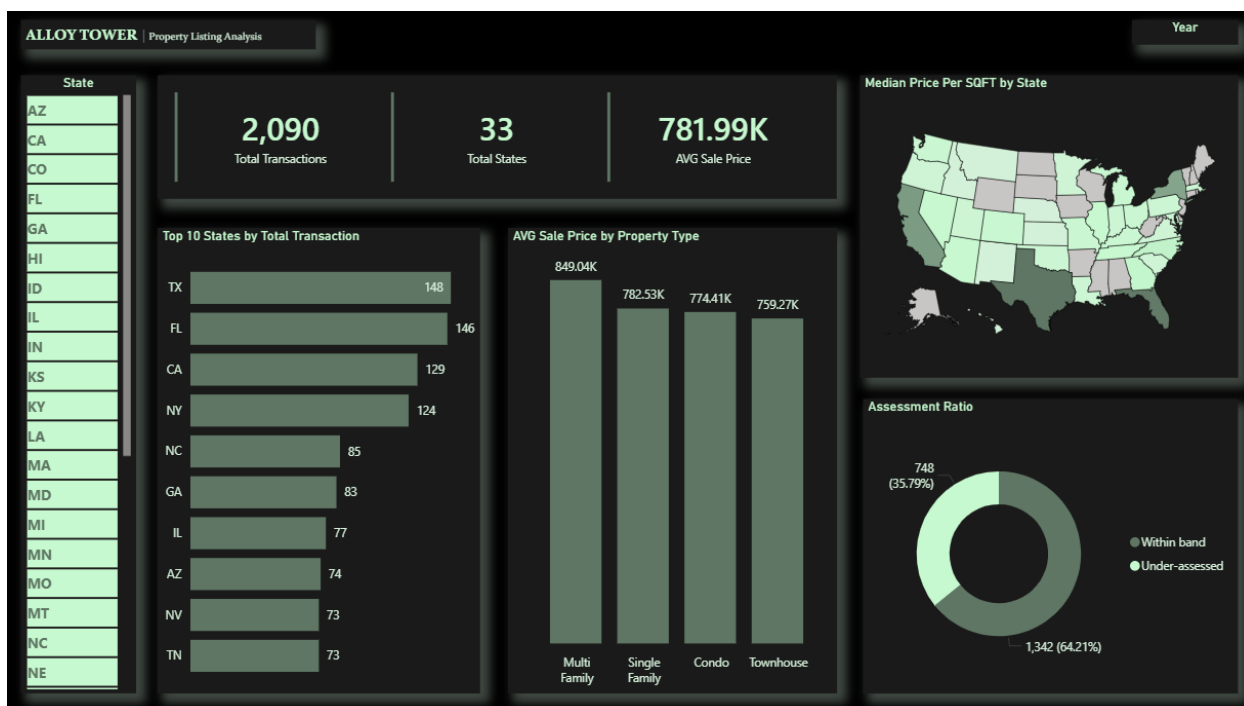
1. **Total Transactions:** Shows the number of property records available, giving an immediate sense of the platform's data volume.
2. **Total States Represented:** Shows geographic pricing patterns. It helps identify high-value and low-value states and gives the business a spatial understanding of the market.
3. **Average Sale Price:** Shows how properties are assessed relative to expected benchmarks. It highlights potential undervaluation, which could be useful for tax strategy or investment decisions.

Together, these KPIs help users understand the size, reach, and overall market context of the data within the centralized platform.



Figure 3: KPI summary — 2,090 Total Transactions | 33 States | \$781.99K Average Sale Price

6. Chart Development



To complement the KPIs, four charts were created to highlight the most meaningful insights in the dataset:

4. **Top 10 States by Total Transactions:** Shows where property activity is most concentrated.
5. **Average Sale Price by Property Type:** Highlights value differences across property categories.
6. **Median Price per Square Foot by State (Map):** Provides a geographic view of pricing patterns.

7. **Assessment Ratio (Donut Chart):** Shows the proportion of under-assessed versus within-band properties.

These visuals were selected because they rely on fields that are complete, consistent, and interpretable, making them suitable for inclusion in a centralized data platform.

7. Slicers — State and Year

Two slicers — State and Year — were added to make the dashboard interactive and flexible. These filters allow users to explore the data by location and time period, supporting deeper analysis and enabling the platform to adapt to different business questions.

8. Dashboard Summary and Insights

The dashboard provides a clear overview of the property data by combining three KPIs, four visual charts, and two slicers to show how the dataset behaves across states and property types. The KPIs highlight the total number of transactions, the number of states represented, and the average sale price, giving an immediate sense of the dataset's size, coverage, and overall market value. The charts then expand on this by showing where transactions are most concentrated, how property types differ in price, how pricing varies geographically, and how properties are assessed relative to expected benchmarks. The State and Year slicers allow users to filter the view by location and time period, making the dashboard interactive and enabling more targeted analysis. Together, these elements give Alloy Tower a realistic understanding of the dataset and demonstrate how the centralized platform can support informed decision-making and future modelling.



Figure 4: AlloyTower Property Listing Analysis Dashboard — Power BI

9. Final Recommendation

Although the dataset was not sufficient to answer every business question, the cleaned data and dashboard provide a strong starting point for AlloyTower's centralized data platform. We recommend:

- Clarify ambiguous fields such as *assessor_id* and the additional *id* column to ensure the data is correctly understood and used across the platform.
- Create a formal data dictionary to support consistent interpretation and improve the reliability of future analysis.
- Expand and standardize data collection to increase the platform's coverage and strengthen its ability to support advanced analytics.
- Implement ongoing data quality checks to maintain accuracy and prevent errors as new data is added to the platform.
- Integrate additional datasets, such as market or demographic information, to enrich the platform and improve the depth of insights.
- Introduce standardized data submission guidelines to ensure all future data enters the platform in a clean, consistent format.
- Develop a phased roadmap for platform expansion to guide AlloyTower's long-term data strategy and support gradual adoption of advanced capabilities.